



Faculty of Engineering and Technology

Department of Computer Science

Business Intelligence - COMP438

Assignment no.1

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a. What specific issue or challenge is being targeted through the use of predictive analytics?

The issue of medication non-adherence in the United States. This is highlighted as a health epidemic leading to premature deaths and staggering financial costs.

b. What does the FICO Medication Adherence Score represent?

The probability of a patient adhering to a prescription for the first year of therapy. It's a scoring model, ranging from 1 to 500, that indicates how likely a patient is to follow their medication regimen.

c. How was the predictive model developed and trained to estimate the FICO Medication Adherence Score (HoH)? Additionally, did the model perform a classification of the score?

Model Development and Training :

- **The score was created using de-identified data from a large pharmacy-benefits manager.**
- **The data sample included over one million patients diagnosed with five chronic diseases (asthma, depression, diabetes, high cholesterol, or hypertension).**
- **Analytic scientists tracked the patterns of patients who filled and refilled prescriptions versus those who didn't.**
- **They identified key variables and developed separate prescription-adherence models for each of the five diseases.**

Model Classification :

The model does not perform a simple classification but rather accurately ranks patients from least to most likely to be adherent. This ranking allows patients to be segmented into deciles. Figure 4 shows patients being grouped into deciles based on their scores.

d. By examining Figure 4 closely, what analytical or modeling technique has been applied to the produced results?

The core technique applied is segmentation. The FICO Score is used to segment individuals into higher- and lower-risk groups (deciles). This segmentation determines the appropriate number and level of interventions to be applied.

e. What are some of the actionable decisions or interventions that were derived from the prediction outcomes?

Sending automated email/text reminders, simplifying drug regimens, offering financial resource information (co-pay cards, subsidized programs), providing instructions in a patient's native language, nurse visits for extreme cases and applying nuanced, segmented tactics and personalized communications plans.